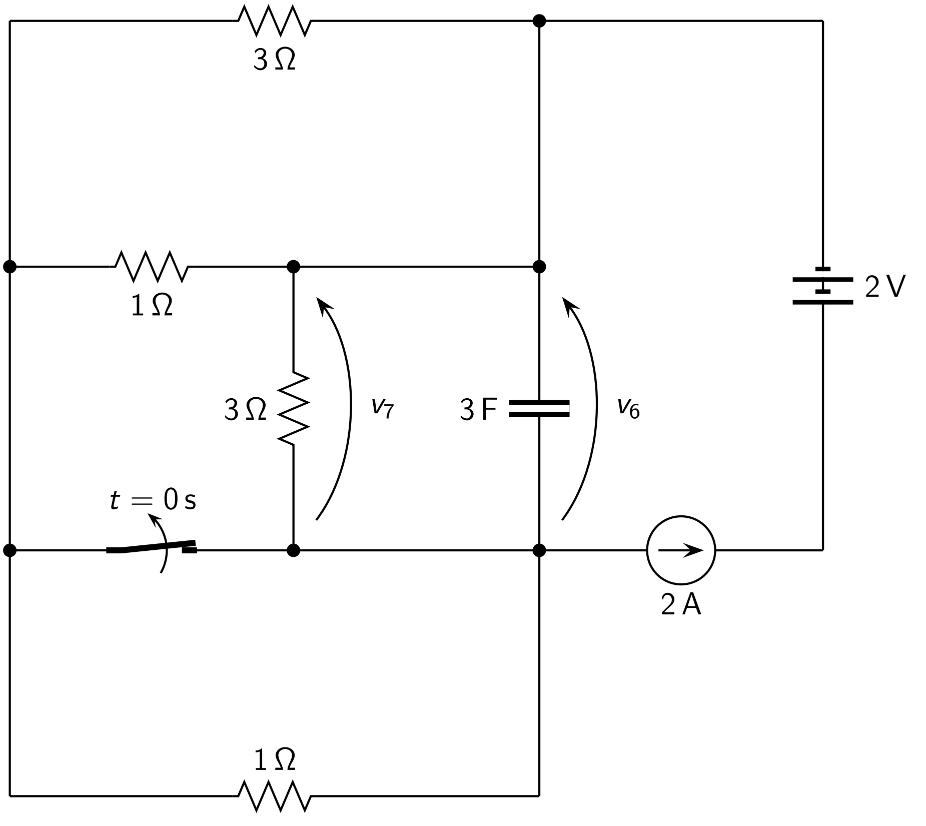
Problem: find the expression of $v_6(t)$, $v_7(t)$ for $t > 0$.



Solution

- Time constant: $\tau = \frac{63}{19}$ s;
- Initial conditions before switching: $v_6(0^-) = \frac{6}{5}$ V;
- Initial conditions after switching: $v_6(0^+) = \frac{6}{5}$ V; $v_7(0^+) = \frac{6}{5}$ V;
- Steady-state solution: $v_6(\infty) = \frac{42}{19}$ V; $v_7(\infty) = \frac{42}{19}$ V;
- Solution for $t > 0$:

$$v_6(t) = \left(\frac{-96}{95}\, e^{-t/\tau} + \frac{42}{19}\right) \text{V}$$

$$v_7(t) = \left(\frac{-96}{95}\, e^{-t/\tau} + \frac{42}{19}\right) \text{V}$$